

Tutorial Letter 102/0/2026

COS2626

Computer Networks I

Year module

School of Computing

This tutorial letter contains the task that must be completed for Assessment
03

BAR CODE

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1. ASSESSMENT OVERVIEW

This practical assessment provides students with an opportunity to apply concepts learned in the module to real-world scenarios, simulations, network analysis, or workplace environments. Students are required to select ONE practical activity from the options below and demonstrate their understanding through practical application, investigation, analysis, or reflection.

The purpose of this assessment is to help you:

- apply theoretical networking knowledge in practical or real-world environments,
- engage with networking tools and technologies,
- develop problem-solving and communication skills,
- and connect networking concepts to South African digital and organizational contexts where appropriate.

This assessment encourages practical exploration, independent learning, and critical reflection.

Regardless of the option selected, all students must demonstrate BOTH:

1. Practical engagement with networking concepts, tools, simulations, configurations, or real-world networking environments; AND
2. Critical explanation or reflection showing understanding of the networking concepts involved.

Students must clearly indicate which domain/topic their task addresses.

2. PRACTICAL TASK OPTIONS

You may select one of the following practical options.

1. Hands-On Projects in Prescribed Book

You may select and execute any one of the Hands-On Projects in the prescribed book.

You must:

- clearly explain the networking concept addressed,
- describe the steps followed,
- indicate the tools/software used,
- include screenshots or evidence of practical work,
- and discuss the outcomes achieved.

2. Network Investigation and Mapping

Investigate and document a home, workplace, or personal network.

Requirements:

- Identify all networking devices.
- Identify connection types used.
- Create a network diagram using software such as draw.io, PowerPoint, Visio, or similar tools.
- Explain how devices communicate within the network.
- Describe the role of each major device.

Evidence Required:

- Network diagram.
- Photographs or screenshots where applicable.

3. Wireless Network Analysis and Security Assessment

Use a Wi-Fi analysis tool or mobile application to examine nearby wireless networks.

Requirements:

- Identify at least five nearby wireless networks.
- Record the SSID, signal strength, channel number, and security type for each network.
- Identify any evidence of channel congestion or wireless interference.
- Evaluate the security of your own wireless network.
- Recommend at least two improvements that could enhance wireless performance or security.
- Interpret the results obtained and explain what they reveal about wireless network performance, coverage, and security.

Evidence Required:

- Screenshots of Wi-Fi analysis results.
- A table summarising the networks identified.
- A discussion of findings and interpretation of results.
- Recommendations for improving wireless network performance and security.

4. Network Diagnostic Testing

Use at least three network diagnostic tools.

Examples:

- ping
- tracert/traceroute
- ipconfig/ifconfig
- nslookup

Requirements:

- Perform diagnostic tests.
- Record command outputs.
- Explain the purpose of each command.
- Interpret the results obtained.
- Discuss any network issues identified.

Evidence Required:

- Screenshots of command outputs.
- Explanation and interpretation.

5: Router Configuration Review

Access and review your wireless router settings if permission is available.

Requirements:

- Record the SSID name.
- Identify encryption type used.
- Determine the number of connected devices.
- Determine whether the default administrator password has been changed.
- Evaluate overall router security.

Important:

Do not disclose passwords or sensitive information.

If access to router settings is unavailable, explain why and perform an alternative network security assessment.

Evidence Required:

- Screenshots where permitted.
- Security evaluation.

6: Packet Analysis using Wireshark

Capture and analyse network traffic.

Requirements:

- Install and use Wireshark.
- Capture traffic while browsing a website.
- Identify at least two networking protocols.
- Explain the role of each protocol.
- Describe the sequence of events that occurs when a webpage loads.

Evidence Required:

- Packet capture screenshots.
- Analysis and interpretation.

7. Network Simulation

Complete a networking simulation using tools such as:

- Cisco Packet Tracer
- GNS3
- Wireshark
- Other approved networking tools

Possible activities:

- Design a small office network.
- Configure DHCP services.
- Troubleshoot connectivity problems.
- Analyse packet transmission.
- Implement subnetting.

Requirements:

- State the objective.
- Describe steps followed.
- Include screenshots.
- Discuss challenges encountered.
- Explain lessons learned.

Evidence Required:

- Screenshots.
- Configuration outputs where applicable

8. Small Business Network Design

Scenario

A small business employs 10 staff members and requires:

- Internet access
- Wireless connectivity
- Shared printing
- Secure access to company resources

Requirements

Design a suitable network solution for the business.

Your design should include:

- Internet connection
- Router or firewall device
- Switch
- Wireless access point(s)
- Staff computers or devices
- Shared network printer
- Appropriate network cabling or wireless connections

You must:

- Create a network diagram.
- Recommend suitable networking hardware.
- Explain the role of each major network device.
- Justify your design decisions.
- Discuss security measures that should be implemented to protect the network.
- Explain how users will access shared resources and the Internet.

Evidence Required

- Network diagram.
- Written explanation and justification of the design.

9. Educational Networking Video

Create a short educational video (5–10 minutes) explaining or demonstrating a networking concept.

Examples include:

- Demonstrating Packet Tracer
- Explaining subnetting
- Demonstrating Wireshark packet analysis
- Configuring a wireless network
- Explaining network security concepts
- Demonstrating troubleshooting techniques

Requirements

- The video must be original.
- Your narration or explanation must be included.
- Visual demonstrations, screenshots, or practical examples should be shown where possible.
- The content should clearly demonstrate your understanding of the networking concept being presented.
- Include a brief reflection on what you learned while preparing the video.

Submit

- A link to the video.
- A written summary (approximately 1–2 pages).

Acceptable Platforms

Videos may be uploaded to:

- YouTube (Unlisted or Public)
- Microsoft OneDrive
- Google Drive
- Any other accessible online platform

Students are responsible for ensuring that the video link is accessible.

3. SUBMISSION INSTRUCTIONS

3.1 Compilation of the document

Submit:

- a written report, OR
- a video link with supporting written summary.

Your submission must include:

- screenshots/evidence where applicable,
- clear explanations,
- references,
- and practical engagement with networking concepts.

3.2 Report structure

You must the following structure:

1. Introduction
2. Selected Topic/Domain
3. Description of the Practical Activity
4. Tools or Technologies Used
5. Steps Followed / Practical Demonstration
6. Screenshots or Evidence
7. Key Findings or Outcomes
8. Reflection and Lessons Learned
9. References

3.3 Submission of assessment

Complete your document assessment in a Microsoft Word document.

- Use 12-point font size (Arial), 1.5 paragraphs spacing on typed A4 pages in Microsoft Word & provide page numbers and section headings when completing the Assessment.
- Save the file as Studentnumber_StudentSurname Initials_COS2626_03_2026, e.g., **1234567_JAMES M R_COS2626_03_2026**.
- Convert the file to PDF.

Once you have the PDF file, do the following on myUNISA:

1. Click on the Assessment 3 link on the left-hand panel or on the Assessment 3 box on the home page.
2. Click on Add Submission
3. Upload your PDF file
4. Click on Save Changes
5. Click on Submit Assignment

4. MARKING GUIDE

The final mark will be awarded holistically based on the overall quality of the practical activity, evidence submitted, understanding demonstrated, analysis provided, and reflection on learning.

Marking Guide:

Excellent	Practical activity completed comprehensively. High-quality evidence provided. Explanations are accurate, detailed, and demonstrate strong understanding of networking concepts. Analysis and reflection are insightful and well-developed.	34–40
Good	Practical activity completed successfully. Sufficient evidence provided. Explanations are mostly accurate and demonstrate good understanding of networking concepts. Analysis and reflection are clear and relevant.	26–33
Satisfactory	Practical activity completed adequately. Evidence provided but may lack detail. Explanations demonstrate basic understanding of networking concepts. Limited analysis or reflection.	20–25
Limited	Practical activity partially completed. Evidence is insufficient or incomplete. Explanations are unclear or demonstrate limited understanding. Minimal analysis or reflection.	10–19
Poor / Inadequate	Very little evidence of practical engagement. Major omissions. Explanations are inaccurate or missing. Reflection absent or inadequate.	0–9

5. TURNITIN & THE USE OF ARTIFICIAL INTELLIGENCE TOOLS

Please take note of the following important information regarding your assessments and how marks will be subtracted for plagiarism and the use of Artificial Intelligence (AI) tools. **The Turnitin Report and AI Report will be generated on the system.** You will be able to see the Turnitin Report but not the AI Report.

5.1 Turnitin

Plagiarism is the act of presenting other people's work as your own. Overt plagiarism can involve getting someone else to do your work for you. But in research, it also means using other people's words or ideas and not acknowledging that you got those ideas from them. The issue of plagiarism is taken very seriously in any field of research. The major mechanism for preventing plagiarism in research is by referencing. In other words, at the end of your research, you specify all the sources that you consulted in a reference list. But this is not enough! In the text of your document, wherever you use the words or ideas of one of those sources, you add a citation. A citation acts like a pointer to one of the sources in the list of references and says to the reader: "This idea is not my own, it was obtained from this source."

Turnitin is the web-based plagiarism detection and prevention tool used by UNISA. Turnitin is software that is often used to detect similarities in research. When a document is submitted to Turnitin, it is compared to a massive database of research documents, and all the parts that are the same or similar, are highlighted. Turnitin awards a similarity index to the document, specifying how similar it is to another research. The best way to lower the similarity index of your documents is to express other researchers' ideas in your own words. Note that this does not mean that you can omit the citation and its corresponding reference. Even though you phrase it in your own words, the idea still belongs to someone else, and so you should add a citation.

Turnitin helps instructors ensure academic integrity by checking submitted assignments for similarity to other sources. Turnitin compares student submissions against a large database of:

- Academic articles
- Student papers (past and present)
- Web content

Turnitin generates a similarity report that highlights matched content and provides a similarity percentage.

5.2 The use of AI tools

Please be reminded that **all assignments must be your own original work**. The use of **Artificial Intelligence (AI) tools**, such as ChatGPT or other content generators, to write or complete your assignments is discouraged.

Students must submit original work. Screenshots, diagrams, packet captures, photographs, and explanations must be generated by the student. The University may request additional evidence or clarification if authenticity is questioned.

Submitting AI-generated content as your own violates the university's academic integrity policy and may be considered plagiarism or misrepresentation. **Turnitin will detect unusual patterns in writing and the use of AI tools.**

Please:

- Use your own knowledge and research to complete tasks.
- Properly paraphrase and cite all sources used.

Please uphold the values of honesty, responsibility, and ethical academic practice.

The following Table provides an outline of the mark deductions for the assessment if your report reveals a high similarity index or the use of AI tools is detected.

Category	Similarity / AI Percentage	Deduction
Turnitin Similarity Report	≤ 25% and similarity acceptable	0
	26% – 35% OR ≤ 25% but similarity not acceptable (e.g., one or more sources ≥ 5%)	-10
	36% – 49%	-20
	50% – 75%	-50
	≥ 76%	Fail module (-100)
AI Usage	≤ 25% and AI usage acceptable	0
	26% – 35% OR ≤ 25% but AI usage not acceptable	-10
	36% – 49%	-20
	50% – 75%	-50
	≥ 76%	Fail module (-100)



6. REFERENCING

Whenever you use/quote research, you must reference it. There are two components of correct referencing technique:

References: All the sources that you consulted in the preparation of the document should have an entry in the list of references at the end of the document. Each entry should have specific information required to identify the source, should the reader wish to find it to check whether it states what you claim it does.

Citations: Wherever a source is used or referred to in the text of your document (particularly in the literature review, but also in any other section) you need to add a citation. A citation is like a pointer to the reference, so that the reader can turn to the list of references and quickly identify the details of the source. In other words, a citation uniquely identifies a source specified in the list of references. Note that every citation in the text of your document should point to some source in the list of references. (In other words, do not cite sources that haven't been referenced.) Also, every source in your list of references should be cited somewhere in the text. (In other words, do not list sources that have not been cited.)

You can also follow the guidelines on: [Introduction to Referencing - CEMS Harvard Referencing Style - LibGuides at University of South Africa \(UNISA\)](#)

7. IN CLOSING

Wishing you all the best for the assessment!!!

Prof H. Abdullah – Project Leader

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